

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB

ASSIGNMENT-4

IFFAH SHAFEEQUE
23MIS0072

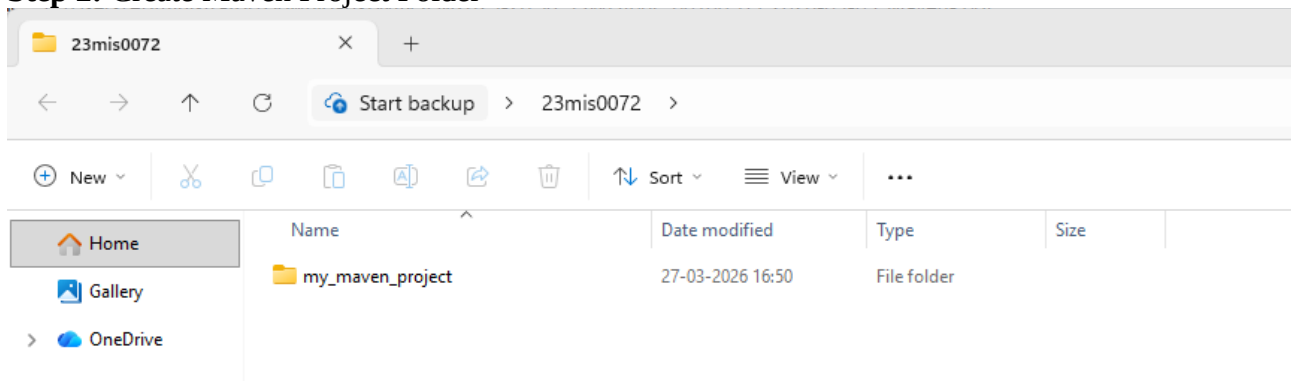
Jenkins Maven Project Tutorial (Without Maven Archetype)

Step 1: Install Required Software

Install the following software: 1. Java Development Kit (JDK) 2. Apache Maven 3. Jenkins 4. Git
Verifying installation:

```
PS C:\Users\Administrator> java -version
java version "21.0.9" 2025-10-21 LTS
Java(TM) SE Runtime Environment (build 21.0.9+7-LTS-338)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.9+7-LTS-338, mixed mode, sharing)
PS C:\Users\Administrator> mvn -version
Apache Maven 3.9.12 (848fbb4bf2d427b72bdb2471c22fced7ebd9a7a1)
Maven home: C:\Program Files\Java\apache-maven-3.9.12
Java version: 21.0.9, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\Administrator> git --version
git version 2.52.0.windows.1
PS C:\Users\Administrator> |
```

Step 2: Create Maven Project Folder



Step 3: Project Structure

Create the following directory structure inside the project folder:

```
my_maven_project/
├── pom.xml
├── src/
│   ├── main/
│   │   ├── java/
│   │   │   ├── com/
│   │   │   │   ├── example/
│   │   │   │   └── App.java
```

Step 4: Create pom.xml

Create a file named pom.xml in the root folder with the following content:

```
pom.xml X
C: > Users > Administrator > Desktop > 23mis0072 > my_maven_project > pom.xml
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4     http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6   <modelVersion>4.0.0</modelVersion>
7
8   <groupId>com.example</groupId>
9   <artifactId>my_maven_project</artifactId>
10  <version>1.0-SNAPSHOT</version>
11
12  <build>
13    <plugins>
14      <plugin>
15        <groupId>org.apache.maven.plugins</groupId>
16        <artifactId>maven-compiler-plugin</artifactId>
17        <version>3.8.1</version>
18        <configuration>
19          <source>1.8</source>
20          <target>1.8</target>
21        </configuration>
22      </plugin>
23    </plugins>
24  </build>
25 </project>
26
```

Step 5: Create Java File

Create the file App.java in the following location: src/main/java/com/example/App.java Add the following code:

```
J App.java X
C: > Users > Administrator > Desktop > 23mis0072 > my_maven_project > src > main > java > com > example > J App.java
1 package com.example;
2 public class App {
3     public static void main(String[] args) {
4         System.out.println("Hello Maven Jenkins Project");
5     }
6 }
7
```

Step 6: Build Maven Project Locally

Open terminal inside the project folder and run:

mvn clean package

```

PS C:\Users\Administrator> cd C:\Users\Administrator\Desktop\23mis0072\my_maven_project
PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:my_maven_project >-----
[INFO] Building my_maven_project 1.0-SNAPSHOT
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom (5.3 kB at 15 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-35.pom

```

Step 7: Build Output

If build is successful you will see:

BUILD SUCCESS

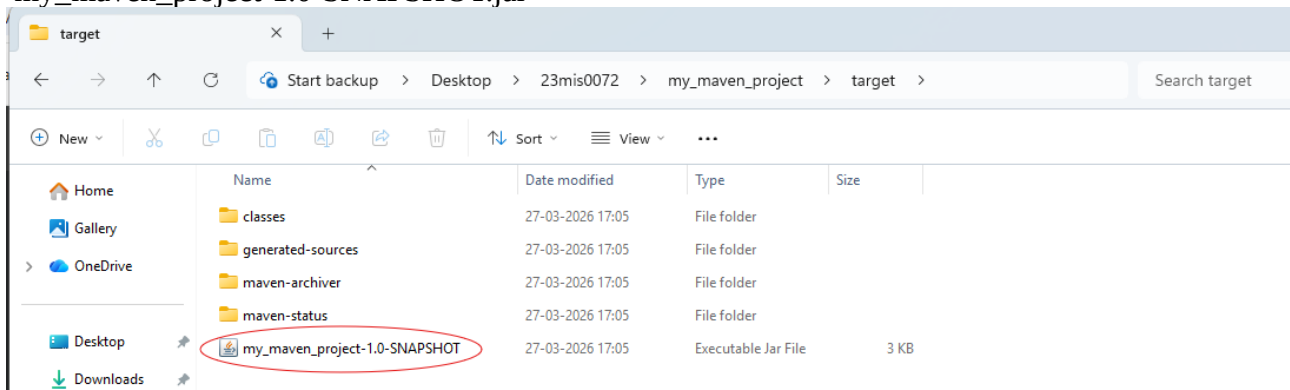
```

[INFO] Building jar: C:\Users\Administrator\Desktop\23mis0072\my_maven_project\target\my_maven_project-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 19.699 s
[INFO] Finished at: 2026-03-27T17:05:58+05:30
[INFO]
PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project>

```

target/

my_maven_project-1.0-SNAPSHOT.jar



Step 8: Push Project to GitHub

Initialize Git repository and push the project:

git init

git add .

git commit -m "Maven Project"

git branch -M main

git remote add origin https://github.com/iffah-shafeeqe/my_maven_project

git push -u origin main

```

PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git init
Initialized empty Git repository in C:/Users/Administrator/Desktop/23mis0072/my_maven_project/.git/
PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git commit -m "Initial Maven Project"
[master (root-commit) 197781f] Initial Maven Project
  Committer: unknown <Administrator@SJT219SITE060.VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

7 files changed, 36 insertions(+)
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/App.java
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/my_maven_project-1.0-SNAPSHOT.jar

```

```

PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git branch -M main

```

```

PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git remote add origin https://github.com/iffah-shafeeque/my_maven_project

```

```

PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 20 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (23/23), 3.27 KiB | 1.09 MiB/s, done.
Total 23 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/iffah-shafeeque/my_maven_project
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Administrator\Desktop\23mis0072\my_maven_project>

```

Step 9: Open Jenkins

Open Jenkins in your browser: <http://localhost:8080>

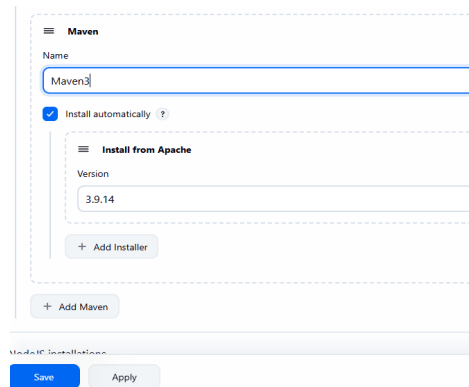
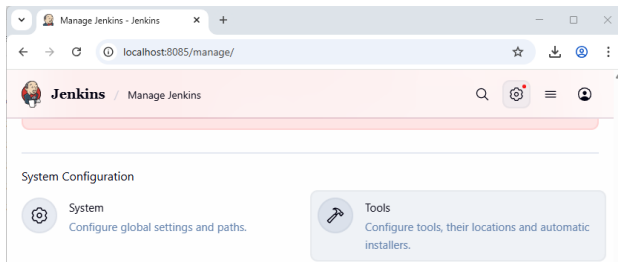
The screenshot shows the Jenkins Dashboard in a web browser. The page title is "Dashboard - Jenkins". The address bar shows "localhost:8080". The dashboard includes a sidebar with links: "New Item", "Build History", "Project Relationship", and "Check File Fingerprint". The main content area shows a table of build history for the "DEV_AQ_Pipeline". The table has columns: "S" (Status), "W" (Icon), "Name", "Last Success", "Last Failure", "Last Duration", "Built On", "Coverage", "F" (Favorite), and "# Issues".

S	W	Name	Last Success	Last Failure	Last Duration	Built On	Coverage	F	# Issues
✓	☁	add	1 mo 12 days #4	1 mo 12 days #3	1.9 sec		N/A	☆	-
✓	☀	DEV_JOB	2 mo 2 days #3	N/A	1.3 sec	Jenkins	N/A	☆	-
✓	☀	dheeraj	N/A	N/A	N/A		N/A	☆	-

Below the table, there is a "Build Queue" section showing "No builds in the queue." and a "Build Executor Status" section showing "(0 of 2 executors busy)".

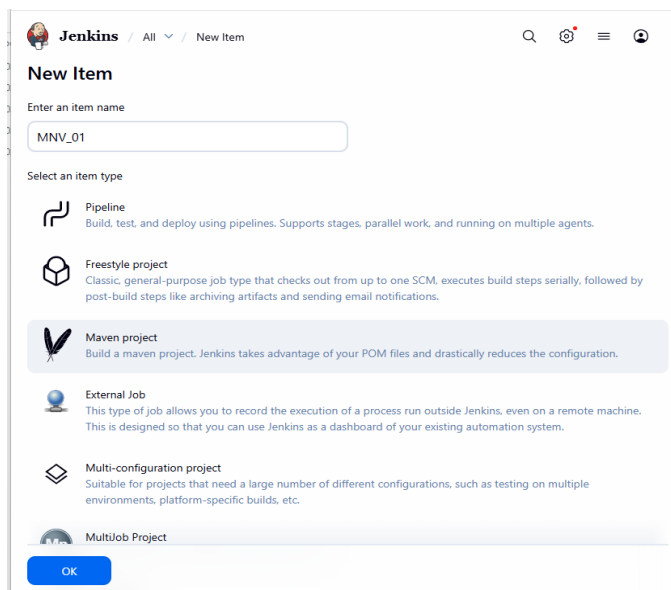
Step 10: Configure Maven in Jenkins

Go to: Manage Jenkins → Global Tool Configuration Add Maven installation: Name: Maven3 Install automatically: ✓



Step 11: Create Maven Project Job

In Jenkins Dashboard: Click New Item Enter Name: MVN_01 Select: Maven Project Click OK



Step 12: Configure GitHub Repository

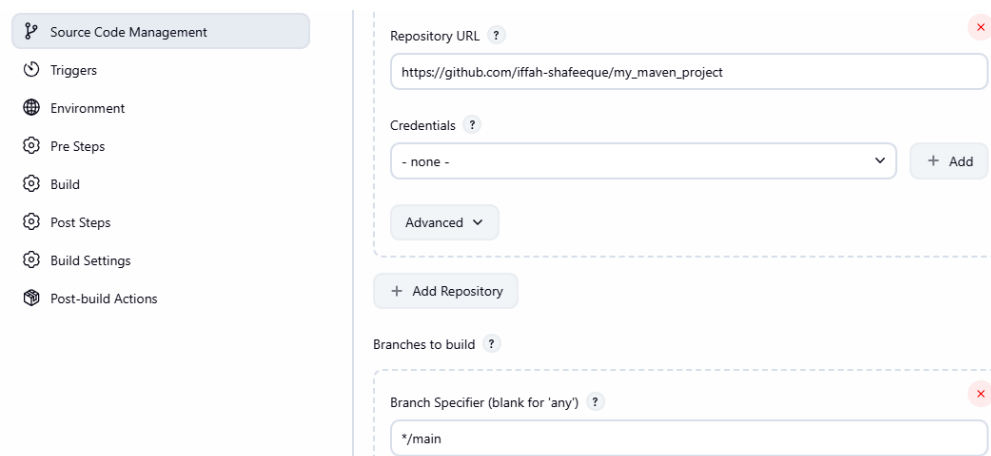
In Source Code Management select Git and provide:

Repository URL:

https://github.com/Ramlingams/my_maven_project.git

Branch:

*/main



Step 13: Configure Maven Build

In Build section provide:

Root POM : pom.xml

Goals and options : clean package

Build

Maven Version

M3

Root POM ?

pom.xml

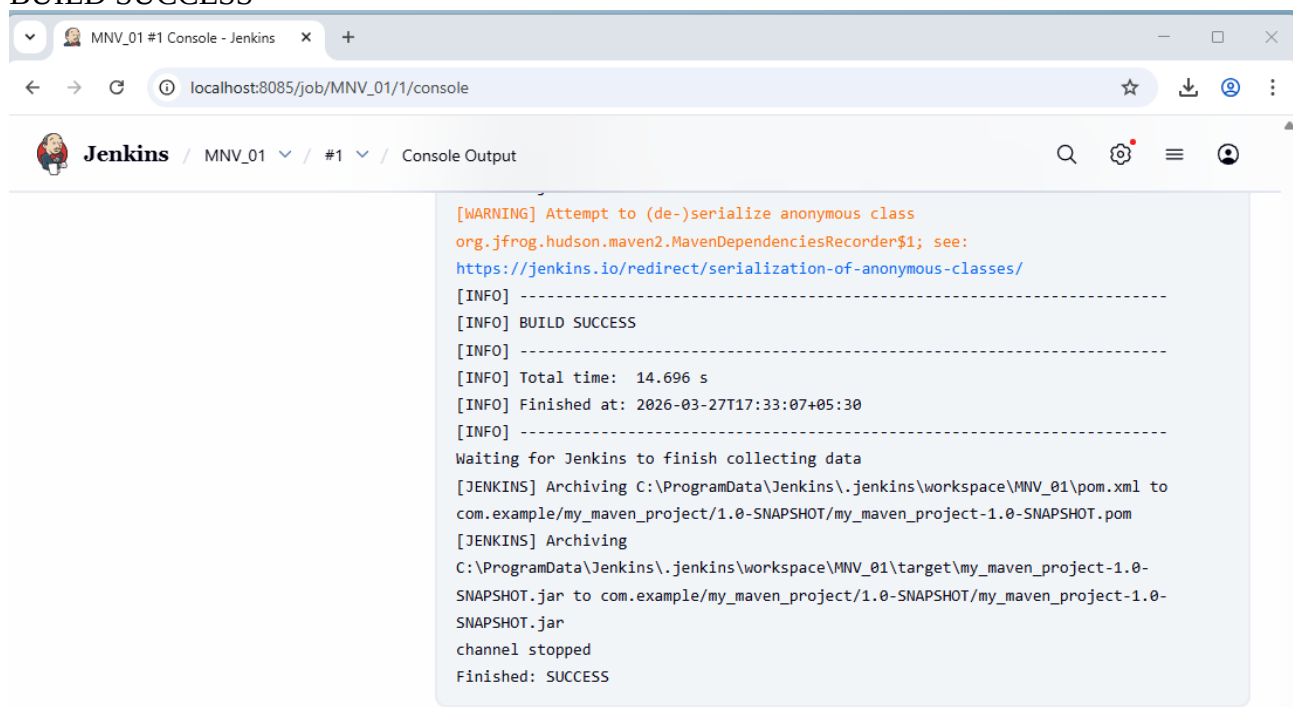
Goals and options ?

clean package

Step 14: Run Jenkins Build

Click Build Now and open Console Output. Expected result:

BUILD SUCCESS



The screenshot shows the Jenkins web interface. The browser address bar indicates the URL is localhost:8085/job/MNV_01/1/console. The Jenkins header shows the job name MNV_01 and the build number #1. The console output displays the following text:

```
[WARNING] Attempt to (de-)serialize anonymous class
org.jfrog.hudson.maven2.MavenDependenciesRecorder$1; see:
https://jenkins.io/redirect/serialization-of-anonymous-classes/
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 14.696 s
[INFO] Finished at: 2026-03-27T17:33:07+05:30
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\MNV_01\pom.xml to
com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-SNAPSHOT.pom
[JENKINS] Archiving
C:\ProgramData\Jenkins\.jenkins\workspace\MNV_01\target\my_maven_project-1.0-
SNAPSHOT.jar to com.example/my_maven_project/1.0-SNAPSHOT/my_maven_project-1.0-
SNAPSHOT.jar
channel stopped
Finished: SUCCESS
```

Step 15: Final Output

The compiled JAR file will be available in Jenkins workspace:

Workspace of MNV_01 on Built-In Node

